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資料結構

Data Structure

Lab 06

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Lab06-Q1&Q2.
Please complete the following sections of the sample code: **– Create a class named 'Queue' that includes the methods: 'enqueue', 'dequeue', 'isEmpty', and 'isFull'.** **– Implement the 'initializeDeck' method.** **– Implement the 'initializePlayer' method.** **– Create the 'playerTurn' method.** **– Create the 'dealerTurn' method.** **Add comments to the program to clearly describe its actions.**

Code

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>
#include <random>
#include <ctime>
using namespace std;
struct Player {// 玩家結構體，包含名稱和分數
    string name; // 玩家名稱
    int score;   // 玩家分數
};
class Card {//建立一個 Card 類別，用來表示單張撲克牌
public:
    string suit; // 花色
    string rank; // 點數
    Card() {}
    Card(string s, string r) : suit(s), rank(r) {}
    void display() const {
        cout << rank << " of " << suit << endl;
    }
};
class Deck {// 牌組類別，負責生成和洗牌
private:
    Card cards[52]; // 52 張牌
    int index; // 目前發牌索引
```

```
public:
    Deck() : index(0) {// 初始化牌組
        string suits[] = { "Hearts", "Diamonds", "Clubs", "Spades" };// 四種花
色
        string ranks[] = { "2", "3", "4", "5", "6", "7", "8", "9", "10", "J",
"Q", "K", "A" };// 13 張牌
        int k = 0;
        for (int i = 0; i < 4; i++) { // 4 種花色
            for (int j = 0; j < 13; j++) { // 13 種點數
                cards[k] = Card(suits[i], ranks[j]);
                k++;
            }
        }
    }
    void shuffleDeck() {// 洗牌
        srand(time(0));
        for (int i = 51; i > 0; i--) {// 從最後一張牌開始，向前遍歷
            int j = rand() % (i + 1);// 生成隨機索引 j
            swap(cards[i], cards[j]);// 交換牌組中的兩張牌
        }
    }
    // 取得指定位置的牌
    Card getCard(int i) const {
        return cards[i];
    }
};
// Queue 類別
class Queue {
private:
    vector<Card> deck; // 用來存放卡牌的容器
    int Front;         // 指向牌堆的前端
    int Rear;          // 指向牌堆的尾端
    int capacity;      // 容器最大容量
public:
    // 建構函式
    Queue(int size) : Front(-1), Rear(-1), capacity(size) {}
    // 插入元素到佇列尾端
    void enqueue(Card card) {
```

```
        if (isFull()) { // 檢查是否滿
            cout << "牌堆已滿，無法插入！" << endl;
            return;
        }
        if (Front == -1) {
            Front = 0;
        }
        deck.push_back(card); // 插入元素到尾端
        Rear = static_cast<int>(deck.size()) - 1; // 更新尾端指標
    }
    // 從佇列中移除最前端元素
    void dequeue() {
        if (isEmpty()) { // 檢查是否空
            cout << "牌堆為空，無法獲取最前端的牌！" << endl;
            return;
        }
        deck.erase(deck.begin()); // 移除最前端元素
        if (deck.empty()) {
            Front = -1;//重置牌堆指標
            Rear = -1;
        } else {
            Rear --; // 更新尾端指標
        }
    }
    Card front() { // 取得最前端元素的值
        if (Front == -1) {
            cout << "排堆為空！" << endl;
            return Card(); // 回傳一個空的Card 對象
        }
        return deck[Front]; // 回傳最前端元素的值
    }
    bool isEmpty() const {// 判斷牌堆是否為空
        return Front == -1;
    }
    bool isFull() const {// 判斷牌堆是否已滿
        return deck.size() == capacity;//當牌堆的牌等於最大容量，表示牌堆已滿，回
傳True
    }
```

```
};
void initializeDeck(Queue& cardDeck) {// 初始化並洗牌
    Deck deck;
    cout << "初始牌堆: \n";
    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display(); //印出原始牌堆
    }
    deck.shuffleDeck();//洗牌
    cout << "\n洗牌後的牌堆: \n";
    for (int i = 0; i < 52; i++) {
        deck.getCard(i).display();//印出洗牌後的牌堆
    }
    //提示:將洗好的牌放入牌堆
    //使用for迴圈，將洗好的牌(deck.getCard(i))放入牌堆
        for (int i = 0; i < 52; i++) {
            cardDeck.enqueue(deck.getCard(i)); //將洗好的牌放入牌堆
        }
}
void initializePlayer(Player* player, string name, Queue& cardDeck) {// 初始化
玩家，發兩張牌
    player->name = name;// 玩家名稱
    player->score = 0;// 玩家分數
    cout << player->name << " 抽到的牌: ";

    for (int i = 0; i < 2; i++) {// 發兩張牌
        Card drawnCard = cardDeck.front(); //從牌堆中取出一張牌
        cardDeck.dequeue(); //從牌堆中移除這張牌

        // 將抽到的牌印出
        cout << drawnCard.rank << " of " << drawnCard.suit << "  ";

        if (drawnCard.rank == "A") player->score += 1;//計算A分數
        else if (drawnCard.rank == "J" || drawnCard.rank == "Q" ||
drawnCard.rank == "K") player->score += 10;//計算JQK分數
        else player->score += stoi(drawnCard.rank);//計算2-10分數
    }
    cout << "\n";
```

```
    cout << player->name << " 玩家的初始分數: " << player->score << endl;//印出玩家的初始分數
}
void playerTurn(Player* player, Queue& cardDeck) {// 玩家回合
    char choice;
    while (player->score < 21 && !cardDeck.isEmpty()) {// 玩家小於 21 點且牌堆不為空
        cout << player->name << " 您的手牌分數目前為: " << player->score << " 要抽牌嗎？(h = 抽, s = 停) : ";//顯示選擇
        cin >> choice;//輸入選擇

        if (choice == 'h') {//如果玩家選擇抽牌
            Card newCard = cardDeck.front();//抽出牌堆中 front 指向的牌
            cardDeck.dequeue();//從牌堆中移除這張牌
            if (newCard.rank == "A") player->score += 1;//計算 A 點數
            else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank == "K") player->score += 10; //計算 JQK 點數
            else player->score += stoi(newCard.rank);//計算 2-10 點數
            //印出抽到的牌和總點數
            cout << player->name << " 抽到: " << newCard.rank << " of " << newCard.suit << " 總分: " << player->score << endl;
            if (player->score > 21) {//如果玩家大於 21 點
                cout << player->name << " 爆掉了！遊戲結束。\n";//玩家爆牌，莊家獲勝
                return;
            }
        }
        else if (choice == 's') {//如果玩家選擇停牌
            cout << player->name << " 選擇停牌，總分: " << player->score << endl;//印出總點數
            break;
        }
        else {
            cout << "請輸入有效選項 (h = 抽, s = 停)！" << endl;//輸入無效
        }
        }
}
void dealerTurn(Player* dealer, Queue& cardDeck) {// 莊家回合
```

```
    while (dealer->score < 17 && !cardDeck.isEmpty()) {// 莊家小於17點且牌堆不為空
        //提示:從牌堆中取出一張牌，然後從牌堆中移除這張牌
        //注意:卡牌變數的類型是Card，並且使用front抽牌
            Card newCard = cardDeck.front();//抽出front指向的牌
            cardDeck.dequeue();//從牌堆移除這張牌
        if (newCard.rank == "A") dealer->score += 1;//計算A點數
        else if (newCard.rank == "J" || newCard.rank == "Q" || newCard.rank == "K") dealer->score += 10;//計算JQK點數
        else dealer->score += stoi(newCard.rank);//計算2-10點數
        //印出抽到的牌和總點數
        cout << "莊家抽到: " << newCard.rank << " of " << newCard.suit << " 莊家目前總分: " << dealer->score << endl;
        if (dealer->score > 21) {//如果莊家大於21點
            cout << "莊家爆了！玩家獲勝！" << endl;//玩家獲勝
            return;
        }
    }
    }
// 判斷勝負
void determineWinner(Player* player, Player* dealer) {
    if (player->score > 21) cout << player->name << " 爆了！莊家獲勝！\n";//玩家爆牌，莊家獲勝
    else if (dealer->score > 21 || player->score > dealer->score)
         cout << player->name << " 贏了！\n";//莊家爆牌或玩家點數大於莊家，玩家獲勝
    else if (player->score == dealer->score) cout << "平手！\n";//莊家與玩家點數相同，平手
    else cout << "莊家贏了！\n";
}

int main() {
    srand(time(0));
    Queue cardDeck(52);//宣告牌堆
    initializeDeck(cardDeck);//初始化牌堆

    Player player, dealer;//宣告並初始化莊家以及玩家
    initializePlayer(&player, "玩家", cardDeck);//初始化玩家手上的牌
```

```
    initializePlayer(&dealer, "莊家", cardDeck);//初始化莊家手上的牌

    playerTurn(&player, cardDeck);
    if (player.score <= 21) {
        cout << "\n 莊家回合...\n";
        dealerTurn(&dealer, cardDeck);
        determineWinner(&player, &dealer);
    }

    return 0;
}
```

result

(a)

```
初始牌堆:
2 of Hearts
3 of Hearts
4 of Hearts
5 of Hearts
6 of Hearts
7 of Hearts
8 of Hearts
9 of Hearts
10 of Hearts
J of Hearts
Q of Hearts
K of Hearts
A of Hearts
2 of Diamonds
3 of Diamonds
4 of Diamonds
5 of Diamonds
6 of Diamonds
7 of Diamonds
8 of Diamonds
9 of Diamonds
10 of Diamonds
J of Diamonds
Q of Diamonds
K of Diamonds
A of Diamonds
2 of Clubs
3 of Clubs
4 of Clubs
5 of Clubs
6 of Clubs
7 of Clubs
8 of Clubs
9 of Clubs
10 of Clubs
J of Clubs
Q of Clubs
K of Clubs
A of Clubs
2 of Spades
3 of Spades
4 of Spades
5 of Spades
6 of Spades
7 of Spades
8 of Spades
9 of Spades
10 of Spades
J of Spades
Q of Spades
K of Spades
A of Spades
```

(b)

```
洗牌後的牌堆:
3 of Hearts
8 of Spades
Q of Spades
K of Spades
5 of Spades
6 of Hearts
2 of Hearts
4 of Diamonds
K of Diamonds
8 of Hearts
5 of Hearts
J of Hearts
3 of Diamonds
J of Spades
10 of Hearts
4 of Hearts
10 of Diamonds
9 of Clubs
8 of Diamonds
A of Spades
2 of Clubs
J of Clubs
Q of Clubs
9 of Hearts
2 of Diamonds
A of Clubs
5 of Clubs
7 of Hearts
7 of Clubs
7 of Spades
K of Hearts
A of Hearts
6 of Diamonds
9 of Diamonds
10 of Spades
A of Diamonds
Q of Diamonds
7 of Diamonds
8 of Clubs
3 of Spades
K of Clubs
J of Diamonds
2 of Spades
4 of Spades
9 of Spades
10 of Clubs
Q of Hearts
6 of Clubs
6 of Spades
4 of Clubs
3 of Clubs
5 of Diamonds
```

(c)

(d)

bash - 1142_DataStructure [Codespaces: turbo telegram] - Visual Studio Code - Brave

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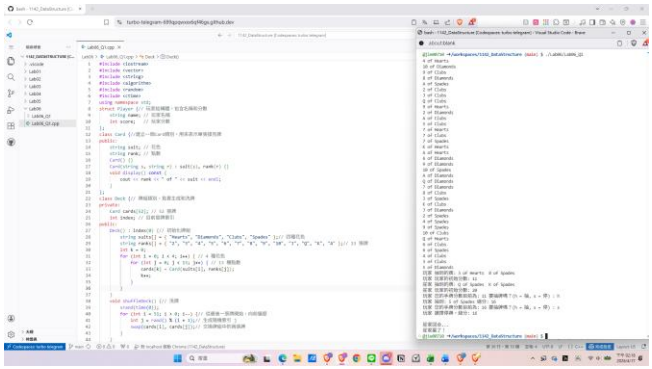
```
@jie00710 ➜/workspaces/1142_DataStructure (main) $ ./Lab06/Lab06_Q1
9 of Spades
10 of Clubs
Q of Hearts
6 of Clubs
6 of Spades
4 of Clubs
3 of Clubs
5 of Diamonds
玩家 抽到的牌: 3 of Hearts  8 of Spades
玩家 玩家的初始分數: 11
莊家 抽到的牌: Q of Spades  K of Spades
莊家 玩家的初始分數: 20
玩家 您的手牌分數目前為: 11 要抽牌嗎？(h = 抽, s = 停) : h
玩家 抽到: 5 of Spades 總分: 16
玩家 您的手牌分數目前為: 16 要抽牌嗎？(h = 抽, s = 停) : s
玩家 選擇停牌，總分: 16

莊家回合...
莊家贏了！
@jie00710 ➜/workspaces/1142_DataStructure (main) $
```



Lab06-Q3
Discussion: The difference between Stack and Queue in terms of card dealing order
Discussion
Queue 為 FIFO，模擬的是「依序取牌」的邏輯，由放進去的第一張牌開始取出， **Stack** 為 GIFO，更貼近現實生活中從「牌堆頂端」拿牌的行為，由最上方的牌(最後放進去的牌)開始抽取。

[EXAMPLE] Q1. Factorial Function

The factorial function n! has value 1 when n<=1 and value n*(n-1)! when n>1.

Please write both a recursive and an iterative C function. Indicate the input, output, and give proper comments. Test your codes and log the results.

Code

<iterative version>

```c
#include<stdio.h>
#include <stdlib.h>
int main()
{
int n,answer;
char i;
printf("Factorial function n!,n=");
scanf("%d",&n); //輸入要查詢的 n
if(n<=1)
{
printf("Answer=1"); //小於 1 時答案為 1
}
else //大於 1 時
{
answer=n; //計算 (n>12 時超出 int 儲存上限)
for(i=(n-1);i>1;i--)
{
answer=answer*i;
}
printf("Answer=%d\n",answer);
}
system("pause");
return 0;
}
```

<recursive version>

...

result

```
Factorial function n!,n=7
Answer=5040
```

Discussion